Education Project Report

Abstract

In this project, I looked at how socioeconomic predictors are related to average ACT scores. I wanted to see if adding state and charter predictors improves model fit. I joined the EdGap dataset and school information for 2016–2017, cleaned the data, and fitted ordinary least squares regression models. The first comparisons used 7,227 schools. The final comparisons used the same 7,028 schools with valid state and charter values. Free or reduced-price lunch had the strongest correlation with scores and the largest absolute standardized coefficient in the reduced model. Adding state increased R-squared from 0.6276 to 0.6776 and reduced mean absolute error from 1.1523 to 1.0369 ACT points. Adding a charter predictor after state made very little difference and was not statistically significant ($p = 0.401$). I chose the model where, in addition to socioeconomic predictors, I added state predictor. These results describe fit to the observed schools.

Introduction

In this project, I looked at how different socioeconomic factors are related to average ACT scores. School averages can help describe differences in academic performance, but they also reflect the circumstances of the students taking the test. I used two datasets: EdGap data and school information for the 2016–2017 school year from the National Center for Education Statistics (NCES, 2019). The ACT variable also includes an ACT-equivalent score when a school reported SAT results. For this project, I wanted to answer four questions: Which socioeconomic factors have the strongest relationship with average ACT scores? Does using multiple socioeconomic factors give a better model than using median income alone? Does adding just a state predictor improve the model? Does the charter predictor still add information after state and the socioeconomic factors are included?

Theoretical Background

I used Pearson correlation to check how strong the relationship is between two variables and whether it is positive or negative. Multiple linear regression helped me look at how several predictors are related to ACT scores at the same time. I also used ordinary least squares (OLS), which finds the best-fitting coefficients by minimizing the sum of squared residuals. Residuals are the differences between actual and predicted ACT scores. These methods can show relationships between variables.

I added state because ACT participation and reporting can be different between states. I also added charter status to see if this will show different results after including socioeconomic factors and state.

Methodology

I performed the analysis in Python in Education.ipynb, using pandas and NumPy for data preparation, Matplotlib and Seaborn for plots, scikit-learn for imputation along with scaling, and statsmodels for

OLS and ANOVA. The input files were EdGap_data.xlsx and ccd_sch_029_1617_w_1a_11212017.csv. Both datasets were provided. The README file provides both dataset download links. To reproduce the notebook, the input and output paths must be changed to your local data folder, right now, it includes paths on my laptop, and the cells must be run in order.

The EdGap file contained 7,986 rows and seven columns. The school information file contained 102,183 rows and 65 columns. I kept school ID, year, state, ZIP code, school type, school level, and charter from the school file for analysis. I renamed columns to lowercase, converted the school ID to an object type, and used a left join on ID to connect the school information and EdGap together. Each row represented one school with a separate column for each variable.

The response was average_act. The five numerical predictors were rate_unemployment, percent_college, percent_married, median_income, and percent_lunch. The first four describe the surrounding census tract: unemployment, adults with a college degree, children in married-couple families, and median household income. The lunch variable describes the proportion of students receiving free or reduced-price lunch at the school. Income was measured in dollars.

I inspected data types, distributions, missing values, and duplicate rows. I changed negative lunch values and ACT values below 1 to missing values and kept rows where school_level was High. There were no duplicate rows in this filtered table. I removed schools with missing ACT scores. I filled these numerical predictor values using IterativeImputer with its default settings, using the five predictors without the response. The cleaned data were exported as education_clean.csv, containing 7,227 schools from 20 states.

I first used correlations, scatterplots, and boxplots to explore the data. Then I fitted an income-only linear model and a quadratic model that also included income squared. The full socioeconomic model included all five numerical predictors. The reduced model retained unemployment, college education, and lunch after income, and the married-family predictor was not significant in the full model. I standardized the three retained predictors with StandardScaler, subtracting their means and dividing by their standard deviations. ACT remained in its original units. This allowed coefficient comparisons on the same predictor scale.

For the final comparison, I kept charter values recorded as Yes or No and rows with a state value. This excluded 199 schools recorded as Not applicable for charter and left 7,028 schools from 19 states. I fitted all three final models on this same subset: the five socioeconomic predictors, those predictors plus C(state), and those predictors plus C(state) and C(charter). I compared R-squared, adjusted R-squared, and mean absolute error (MAE). R-squared measures the proportion of score variation accounted for by the fitted model. Adjusted R-squared accounts for the number of fitted terms. MAE is the average absolute difference between observed as well as fitted scores, measured in ACT points. I used ANOVA to test added terms and inspected residual plots. All reported errors were calculated on the data used to fit each model.

Computational Results

I started by comparing the linear and quadratic models using median income as a predictor. The linear model had an R^2 of 0.211 and MAE of 1.713. The quadratic model had a slightly higher R^2 of 0.219 and a lower MAE of 1.697.

Then I used all five socioeconomic predictors in a multiple regression model. This model had an R^2 of 0.6280 and MAE of 1.1453. After removing median income and percent married, the reduced model had an R^2 of 0.6279 and MAE of 1.1455. The ANOVA comparison between these models gave a p-value of 0.807.

I also standardized the predictors in the reduced model to compare their coefficients. Free or reduced-price lunch had the largest coefficient in absolute value at -1.777, followed by percent college at 0.283 and unemployment rate at -0.123.

For the final comparison, I used 7,028 schools with available state and charter information. I compared three regression models.

After adding state, R^2 increased from 0.6276 to 0.6776, and MAE decreased from 1.1523 to 1.0369. Adding charter status changed R^2 to 0.6777, while MAE stayed almost the same.

I also used ANOVA to compare the models. Adding state gave a p-value of approximately 2.90e-203. Adding charter after state gave a p-value of 0.401.

Discussion

The results show that socioeconomic predictors are related to average ACT scores in schools from the dataset. Free or reduced-price lunch has the strongest correlation with scores and the largest absolute standardized coefficient in the reduced model. Holding the other two reduced-model predictors constant, a one-standard-deviation increase in the lunch proportion is associated with about 1.78 fewer ACT points. This describes a relationship with economic disadvantage; it does not show that receiving lunch assistance lowers scores.

Using several socioeconomic factors gives a much better fit than using median income alone. On the same 7,227 schools, R-squared increases from 0.2112 to 0.6280 and MAE drops from 1.7129 to 1.1453 ACT points. The quadratic income model improves fit only slightly. Its small p-value shows that a detectable difference can still be a small improvement in practical terms. The reduced socioeconomic model gives almost the same results as the full model, and the ANOVA result does not provide

[105]:

	R-squared	Adjusted R-squared	Mean Absolute Error
Socioeconomic	0.6276	0.6274	1.1523
Socioeconomic + state	0.6776	0.6766	1.0369
Socioeconomic + charter + state	0.6777	0.6766	1.0369

evidence that income and the married-family predictor add information once the other three factors are

included. This does not mean income is unrelated to achievement: the predictors can contain overlapping information.

Adding the state predictor increases R-squared from 0.6276 to 0.6776 and reduces MAE from 1.1523 to 1.0369 ACT points. Adding the charter predictor after state increases R-squared only slightly, to 0.6777, while adjusted R-squared and MAE stay almost the same. The charter predictor is not statistically significant after including the other factors ($p = 0.401$). Its confidence interval includes zero. I would choose the socioeconomic plus state model as my final model because it gives almost the same results with fewer predictors. State may capture testing differences and other unmeasured differences between states; these results do not identify their separate contributions.

There are limitations. The final data cover 19 states and only 170 charter schools, so the results do not represent all U.S. schools. The initial and final model groups use different samples and should be compared within their own tables. Some predictor values were imputed, and the reported standard errors do not include imputation uncertainty. School averages also depend on who takes the test, and ACT-equivalent SAT scores add measurement uncertainty.

Conclusions

This project examined whether school average ACT scores could be accounted for by socioeconomic information and whether state and charter add useful information. Several socioeconomic factors fit the observed scores much better than income alone. Lunch had the largest absolute standardized coefficient in the reduced model, and state added a clear improvement in fit. Charter added very little after the other factors were included.

I would use the socioeconomic plus state model as the final model among those compared.

References

Fischer, B. (n.d.). *EdGap_data.xlsx* [Dataset]. GitHub. https://github.com/brian-fischer/DATA-5100/blob/main/EdGap_data.xlsx

National Center for Education Statistics. (n.d.). *Common Core of Data (CCD): Public School Directory, 2016–2017*[Dataset]. https://www.dropbox.com/scl/fi/fkafjk8902sq8ptxh94r2/ccd_sch_029_1617_w_1a_11212017.csv?rlkey=gucrdz5f6e38bezz2y3yalxbw&dl=0